# RehabSense: A Low-Cost Bilateral Wearable System with Mounting-Invariant Activity Recognition for ACL Rehabilitation Monitoring

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***Abstract*—Recovery after anterior cruciate ligament (ACL) reconstruction is judged largely by between-limb symmetry, yet objective movement assessment remains confined to laboratories, and low-cost wearables fail when patients re-attach the straps at a different orientation. We present RehabSense, a research platform built around a US$18 wearable in which one ESP32 samples an MPU6050 inertial measurement unit on each shank and two heel force sensors on a single 100 Hz clock. We formulate monitoring as estimation under unknown per-wearing sensor rotations and contribute a calibration that recovers each sensor's bias, neutral gravity direction and rotation axis; a representation that is provably invariant to sensor rotation yet retains posture through the calibrated neutral pose; and a model-selection rule fixed before any results were seen. Under subject-independent evaluation on real bilateral public data, the deployed random forest reaches a macro-F1 of 0.921 that is unchanged under random re-mounting, whereas raw-axis features collapse from 0.943 to 0.199; inference takes 12.9 ms per window (95th percentile). Device-format replay reproduces the training pipeline on 98.4% of windows, transparent asymmetry and quality metrics respond monotonically to injected deficits in simulation, and a cross-dataset test (macro-F1 0.15–0.24) quantifies the domain gap the forthcoming on-device pilot must close.**

***Index Terms*—anterior cruciate ligament, wearable sensors, inertial measurement unit, human activity recognition, orientation invariance, limb symmetry, random forest, Internet of Things**

## I. Introduction

Return to sport after ACL reconstruction is a symmetry decision: patients who returned before passing objective criteria, including a limb symmetry index of at least 90%, were reinjured far more often [1]. Yet symmetry is measured objectively only in laboratories [2], so most of a recovery lasting nine months or more goes unobserved. Remote programmes already match brochure-based home rehabilitation on early outcomes [3], but they rarely measure how the patient moves.

Inertial measurement units (IMUs) are the most common portable modality for ACL assessment, yet most systems remain at technology readiness levels 3–6 [2], [4]. We identify three obstacles. (i) *Mounting*: estimators and activity models that assume a fixed sensor orientation degrade when the strap is re-attached, which at home happens every session [5], [6]. (ii) *Evaluation*: record-wise splits inflate accuracy [7], and public-dataset scores are presented as if they transfer to new hardware despite well-documented domain shift [8], [9]. (iii) *Transparency*: composite "recovery scores" combine components with unexplained weights.

RehabSense is designed around these obstacles. Its contributions are:

1) a low-cost bilateral node in which one ESP32 reads both shank IMUs and the force sensors in the same 100 Hz tick, with an integrity layer that makes loss, jitter, drift and sensor failure explicit rather than imputed;
2) a per-wearing calibration and a feature representation that is provably invariant to sensor rotation while retaining posture through the calibrated neutral pose;
3) a subject-independent benchmark over three public datasets with device-specific robustness tests, and a model-selection rule fixed before results were observed;
4) transparent asymmetry and Movement Quality Index (MQI) metrics with explicit confidence that never impute missing inputs.

## II. Related Work and Research Gap

### A. Wearable Sensing after ACL Reconstruction

Reviews of 49 out-of-lab ACL studies [2] and 32 wearable studies [4] found few systems validated outside the laboratory and none integrated into care. Phan et al. [10] tracked 26 patients with five IMUs at home and found inter-limb knee-extension asymmetry falling from 7.7° to 2.6° between three and nine months, a change the laboratory captured only partly.

TABLE I
REPRESENTATIVE APPROACHES AND THE GAP REHABSENSE ADDRESSES

| Approach | Cost | Out-of-lab | Sync. bilat. | Mount. robust | Activity model† | Transp. metrics |
|---|---|---|---|---|---|---|
| Lab mocap / force plates [2] | High | ✗ | ✓ | – | ✗ | ✓ |
| Free-living multi-IMU [10] | Med. | ✓ | ✓ | ∼ | ✗ | ∼ |
| Low-cost knee wearables [12]–[14] | Low | ✓ | ∼ | ∼ | ✗ | ∼ |
| Invariant HAR [5], [6], [15] | – | – | ✗ | ✓ | ✓ | ✗ |
| **RehabSense (ours)** | **Low** | ✓ | ✓ | ✓ | ✓ | ✓ |

✓ addressed; ∼ partial or system-dependent; ✗ not addressed; – not applicable. †Learned activity model evaluated subject-independently.

Seel et al. [11] estimated knee angles without assuming sensor orientation by identifying the joint axis from motion. Low-cost ESP32 designs estimate knee flexion from thigh and shank MPU6050s [12], add force sensors and a cloud back end for symmetry indices [13], or fuse IMUs at the edge with 3.1–6.4° RMSE against video [14].

### B. Orientation- and Domain-Robust Activity Recognition

Yurtman and Barshan [5] removed absolute orientation by transforming raw signals; Gil-Martín et al. [15] mapped signals into a consistent reference frame; Barua et al. [16] used heuristic invariant features; and Baek et al. [6], motivated by at-home re-attachment, built invariance to per-location rotations into an SO(3)-equivariant network. Invariance, however, discards the segment's posture, which is what separates sitting from standing. Cross-dataset generalisation remains hard even with self-supervised pretraining [8], [9].

### C. Research Gap

Low-cost knee wearables concentrate on single-joint angles, and orientation-robust activity recognition lacks rehabilitation metrics (Table I). To our knowledge, no low-cost system combines (a) synchronous bilateral capture, (b) mounting-invariant activity recognition that keeps posture, (c) subject-independent, pre-registered evaluation that measures its own domain gap, and (d) movement metrics whose components and missing inputs are explicit.

## III. SYSTEM DESIGN

### A. Wearable Node

An ESP32 DevKit V1 reads two GY-521 MPU6050 IMUs, strapped to the lateral left and right shanks, over one $I^2C$ bus at 400 kHz; the AD0 pin selects address 0x68 (left) or 0x69 (right) (Fig. 2). Each IMU is set to $\pm 4$ g and $\pm 500$°/s with a 44 Hz low-pass filter, verified by register read-back at boot. Two FSR402 heel sensors form 10 kΩ dividers on ADC1 pins (ADC2 is unusable with Wi-Fi active); their output is an uncalibrated *load proxy*, never reported as newtons. An 18650 cell feeds a TP4056 charger and an MT3608 5 V boost converter; the parts cost about INR 1,560 (US$18).

A FreeRTOS task on core 1 reads both IMUs and the force channels in the same tick, stamps them with one microsecond timestamp and queues them in a lock-free ring buffer; the main loop sends ten samples per WebSocket frame (optionally over TLS). A failed read is sent as `null`, never as a stale or other-side value, and status frames count ring overflows and $I^2C$ errors to separate device-side from network loss.

**Algorithm 1** Online processing of one device stream

**Input:** stream $(t_k, \mathbf{x}_k^{\mathrm{L}}, \mathbf{x}_k^{\mathrm{R}}, f_k)$; calibration $(\mathbf{b}_\omega, \kappa, \mathbf{g}_0, \mathbf{u})$ per IMU; model $M$ with $W$ and $\tau$

**for** each received packet **do**
  check order, duplicates and gaps; mask invalid values as NaN
  correct bias and scale; update the tilt $\theta_t$ (Eq. 6)
  **if** a stride $s$ has elapsed **then**
    $X \leftarrow$ last $W$ seconds, anti-aliased and resampled to 25 Hz
    **if** more than 10% of $X$ is missing **then**
      emit INSUFFICIENT_DATA
    **else**
      $\boldsymbol{\phi} \leftarrow$ 324 invariant features of $X$ (Eqs. 3–4)
      $\mathbf{p} \leftarrow M(\boldsymbol{\phi})$
      emit $\arg\max \mathbf{p}$ if $\max \mathbf{p} \geq \tau$, else LOW_CONFIDENCE
      emit $A$ (Eq. 8) or SINGLE_SIDE
    **end if**
  **end if**
  on a completed repetition: ROM proxy, SPARC, MQI (Eq. 9)
  persist results with provenance; publish to the dashboard
**end for**

### B. Protocol and Software Platform

A *hello* frame declares each IMU (side, placement, ranges), each force channel and a per-device key. Provenance is fixed at this handshake: a registered device with its own key is LIVE, the simulator is SIMULATED, anything else is UNVERIFIED, and only LIVE data can be used for training. The service (Fig. 1) runs FastAPI and a Next.js dashboard on Vercel over PostgreSQL, whose LISTEN/NOTIFY relays live messages between instances; raw samples (19 MB/h) are kept for 30 days unless consent is given, and the code has 279 back-end and 8 ML tests.

## IV. METHODOLOGY

### A. Problem Formulation

Sample $k$ carries one device timestamp $t_k$ and, for each side $s \in \{\mathrm{L}, \mathrm{R}\}$, a measurement $\mathbf{x}_k^s = [\mathbf{a}_k^s; \boldsymbol{\omega}_k^s] \in \mathbb{R}^6$ (g, °/s), plus force proxies $f_k$. Every wearing applies an unknown rotation $\mathbf{R}^s \in SO(3)$ to each sensor, and each MPU6050 adds its own bias and scale error. For every window the system outputs (i) an activity $y \in \mathcal{Y}$ ($|\mathcal{Y}| = 9$) or an abstention, (ii) per repetition a range-of-motion (ROM) proxy, (iii) an asymmetry score $A \in [0, 1]$ and (iv) an MQI in $[0, 100]$, each with a status and a confidence. Three requirements apply: outputs are invariant to $\mathbf{R}^s$; a missing sensor or side is never imputed or mirrored (only short gaps inside a window are interpolated); and the device path reproduces the training path. Algorithm 1 summarises the online procedure.

### B. Stream Integrity

The sampling rate is measured, never trusted: $\hat{f} = 1/\operatorname{median}(\Delta t)$, while the effective rate $N_{\mathrm{rx}}/(t_{\mathrm{last}} - t_{\mathrm{first}})$ also counts the losses exposed by sequence-number gaps. An IMU is frozen after 25 identical readings and saturated at $\geq 98\%$ of

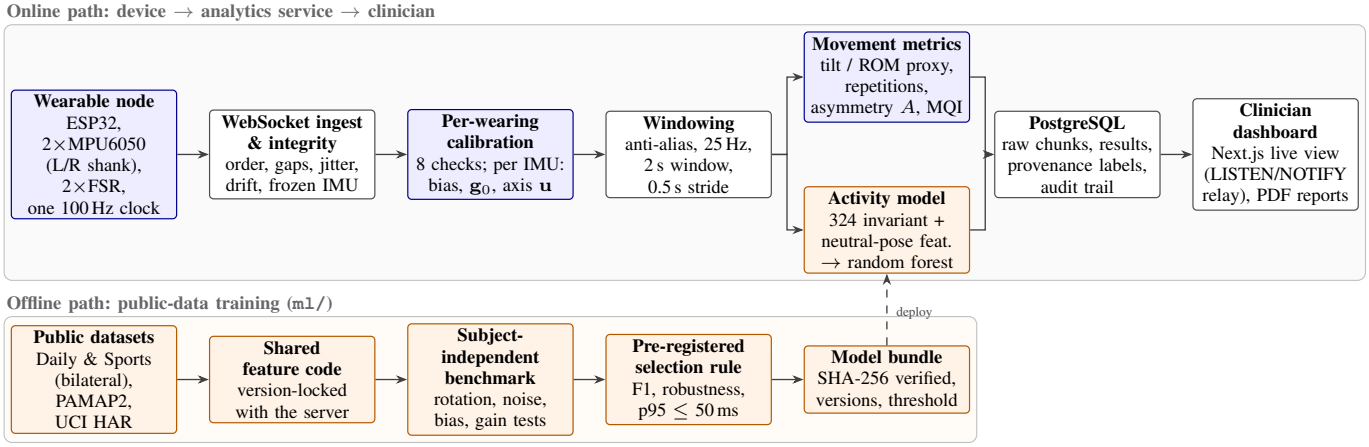



Fig. 1. RehabSense architecture. Online: the node streams synchronised bilateral samples; the service checks stream integrity, calibrates every wearing, windows the signals and runs the activity model and movement metrics; results reach the dashboard through a PostgreSQL relay. Offline: models are benchmarked on public data with the same feature code the server runs, selected by a pre-registered rule and deployed as hash-verified bundles.

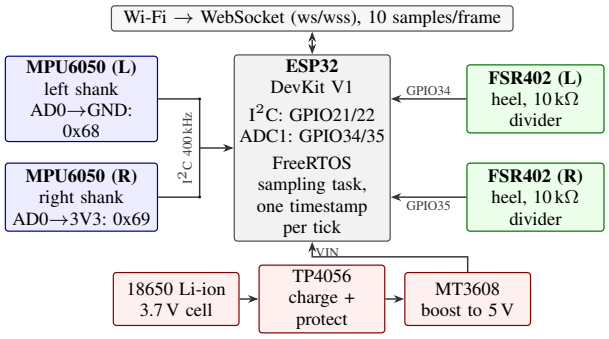



Fig. 2. Wearable node: one ESP32 samples both shank IMUs (shared $I^2C$ bus, distinct addresses) and two heel force sensors in the same tick.

TABLE II
PER-WEARING CALIBRATION CHECKS (THRESHOLDS FOR MPU6050-CLASS SENSORS)

| Check | PASS | WARN (else FAIL) |
|---|---|---|
| 1 IMU connectivity | $\geq 95\%$ valid, not frozen | – |
| 2 Sampling rate | $\lvert\hat{f} - f_{\mathrm{decl}}\rvert / f_{\mathrm{decl}} \leq 5\%$ | $\leq 15\%$ |
| 3 Gravity magnitude | $\lVert\bar{\mathbf{a}}\rVert \in [0.9, 1.1]$ g | $[0.8, 1.2]$ g |
| 4 Accel. baseline | max. SD $\leq 0.02$ g | $\leq 0.05$ g |
| 5 Gyro. baseline | noise $\leq 1°$/s, $\lvert\mathbf{b}_\omega\rvert \leq 20°$/s | noise $\leq 3°$/s |
| 6 Force channel | load response‡ | no load response |
| 7 Neutral pose | $\mathbf{g}_0$ stored | – |
| 8 Baseline movement | RMS $\geq 6°$/s, $\mathbf{u}$ found | axis learned later |

‡SD during movement $> \max(3\sigma_{\mathrm{rest}}, 0.01)$; FAIL if the rest value is $\geq 0.98$ of full scale.

full scale; invalid samples are masked, not repaired. For clock drift, in each 5 s bucket $b$ we keep the packet with the smallest delay $t^{\mathrm{arr}} - t^{\mathrm{dev}}$ and fit

$$t_b^{\mathrm{arr}} = \beta\, t_b^{\mathrm{dev}} + c, \qquad \delta = (\beta - 1) \cdot 10^6 \text{ ppm} \tag{1}$$

by least squares over at least 30 s. This *lower envelope* keeps network queuing from appearing as drift; $|\delta| > 2000$ ppm flags a non-real-time stream.

### C. Per-Wearing Calibration

Stillness is detected, not assumed: the server waits for a 3 s span $\mathcal{S}$ in which, on every healthy IMU, the largest per-axis standard deviation is below 0.03 g and 2.5°/s (timeout 20 s). From $\mathcal{S}$ it estimates, per sensor, the gyroscope bias, the neutral gravity direction and the accelerometer scale,

$$\mathbf{b}_\omega = \bar{\boldsymbol{\omega}}_{\mathcal{S}}, \qquad \mathbf{g}_0 = \bar{\mathbf{a}}_{\mathcal{S}} / \lVert\bar{\mathbf{a}}_{\mathcal{S}}\rVert, \qquad \kappa = 1/\lVert\bar{\mathbf{a}}_{\mathcal{S}}\rVert. \tag{2}$$

During 5 s of slow repetitions $\mathcal{M}$, the dominant rotation axis is the first right-singular vector of the bias-corrected angular velocities, $\boldsymbol{\Omega}_{\mathcal{M}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, $\mathbf{u} = \mathbf{v}_1$, provided their RMS exceeds 6°/s; its sign is fixed by the direction of the first movement so it is stable across sessions. Table II lists the eight checks. The calibration quality $q_{\mathrm{cal}}$ is the mean check score (PASS 1, WARN 0.5, FAIL 0); $\mathcal{S}$ and $\mathcal{M}$ are stored so a calibration can be recomputed from raw data.

### D. Mounting-Invariant Representation

After bias and scale correction, each window is anti-aliased with a moving average and linearly resampled to 25 Hz by one function shared by training and deployment, with $W = 2$ s and stride $s = 0.5$ s (response delay $\approx W/2 + s = 1.5$ s). With the window's gravity direction $\hat{\mathbf{g}} = \bar{\mathbf{a}}/\lVert\bar{\mathbf{a}}\rVert$, each IMU is mapped to six channels:

$$\begin{aligned} &\lVert\mathbf{a}_t\rVert, \quad \mathbf{a}_t \cdot \hat{\mathbf{g}}, \quad \lVert\mathbf{a}_t - (\mathbf{a}_t \cdot \hat{\mathbf{g}})\hat{\mathbf{g}}\rVert, \\ &\lVert\boldsymbol{\omega}_t\rVert, \quad \boldsymbol{\omega}_t \cdot \hat{\mathbf{g}}, \quad \lVert\boldsymbol{\omega}_t - (\boldsymbol{\omega}_t \cdot \hat{\mathbf{g}})\hat{\mathbf{g}}\rVert. \end{aligned} \tag{3}$$

These discard posture, the cue that separates sitting from standing. We restore it relative to the calibrated $\mathbf{g}_0$, measured in the same sensor frame:

$$\theta_t^n = \arccos(\mathbf{a}_t \cdot \mathbf{g}_0 / \lVert\mathbf{a}_t\rVert)\,. \tag{4}$$

*Proposition 1:* For any rotation $\mathbf{R} \in SO(3)$ of a sensor that is fixed within a window, the channels (3) and the neutral angle (4) are unchanged.

*Proof:* The rotation maps $\mathbf{a}_t \mapsto \mathbf{R}\mathbf{a}_t$ and $\boldsymbol{\omega}_t \mapsto \mathbf{R}\boldsymbol{\omega}_t$; by linearity of the mean, $\hat{\mathbf{g}} \mapsto \mathbf{R}\hat{\mathbf{g}}$; and $\mathbf{g}_0$, measured by the same

sensor, maps to $\mathbf{R}\mathbf{g}_0$. Since $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$, $(\mathbf{R}\mathbf{a}) \cdot (\mathbf{R}\hat{\mathbf{g}}) = \mathbf{a} \cdot \hat{\mathbf{g}}$ and $\|\mathbf{R}\mathbf{v}\| = \|\mathbf{v}\|$ for every $\mathbf{v}$, including $\mathbf{v} = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{g}})\hat{\mathbf{g}}$. The same holds for $\boldsymbol{\omega}$ and for $\mathbf{a} \cdot \mathbf{g}_0$. ∎

Each channel yields 24 statistics (moments, percentiles, RMS, skewness, kurtosis, zero-crossing rate, dominant frequency, spectral entropy, four band powers, autocorrelation peak and lag) plus jerk RMS; posture adds eight (statistics of $\theta^n$, acceleration along and rotation about $\mathbf{g}_0$); and cross-side correlation, lag and variance-ratio features add 18, giving 153 features per IMU and 324 bilateral, all invariant by the proposition.

### *E. Activity Recognition and Pre-Registered Selection*

The nine classes are those the public datasets share (lying, sitting, standing, walking, stairs up/down, running, cycling, other exercise); there is no squat class because no public dataset contains one, and the prescribed exercise is known from the session, so the classifier supplies free-living context. Feature models are a random forest (RF; 200 trees, $\sqrt{d}$ features per split, minimum leaf size 2, class-balanced bootstrap) [17], [18], histogram gradient boosting (HGB), and the RF trained with device-like augmentation ($2\times$ noise, $\pm 2°$/s bias, $\pm 3\%$ gain). Deep models encode each IMU's nine-channel sequence (Eq. 3 plus three neutral-pose channels) with a shared three-layer 1D CNN, optionally followed by an LSTM [19], and fuse the embeddings symmetrically as $\mathbf{z} = [\mathbf{h}_\mathrm{L}, \mathbf{h}_\mathrm{R}, |\mathbf{h}_\mathrm{L} - \mathbf{h}_\mathrm{R}|, \mathbf{h}_\mathrm{L} \odot \mathbf{h}_\mathrm{R}]$, so the prediction is independent of which leg wears which sensor (legs are also swapped with probability 0.5 in training; 58.5k/91.8k parameters). They use class-weighted cross-entropy, AdamW ($2 \times 10^{-3}$, cosine schedule, $\leq$ 15 epochs) and early stopping on a held-out training subject; one variant pretrains the encoder on PAMAP2.

The selection rule was fixed before any result was inspected. For each family $m$ with windows $W \in \{1, 2, 3\}$ s,

$$W_m^* = \min\{W : \mathrm{F1}_m(W) \geq \max_{W'} \mathrm{F1}_m(W') - 0.01\},$$
$$m^* = \arg\max_m \mathrm{F1}_m(W_m^*) \;\; \text{s.t.} \;\; \Delta_m^{\mathrm{rot}} \leq 0.03,\; L_m^{95} \leq 50\,\mathrm{ms}, \tag{5}$$

where $\Delta^{\mathrm{rot}}$ is the macro-F1 drop under random rotation and $L^{95}$ the 95th-percentile latency; ties within 0.01 go to the smaller model, and a deep model is adopted only if it beats $m^*$ by $\geq 0.02$ under the same constraints. At run time the model abstains (LOW_CONFIDENCE) when $\max_c p(c \mid \boldsymbol{\phi}) < \tau = 0.3$.

### *F. Segment Tilt and Range-of-Motion Proxy*

With one IMU per leg there is no knee angle, so we estimate a segment tilt about the calibrated axis $\mathbf{u}$, an idea related to the joint-axis identification of [11]. The accelerometer tilt $\theta_t^a$ is the signed angle between $\mathbf{g}_0$ and $\mathbf{a}_t$ projected onto the plane normal to $\mathbf{u}$, fused by a complementary filter [20]:

$$\theta_t = \alpha\,(\theta_{t-1} + (\boldsymbol{\omega}_t \cdot \mathbf{u})\,\Delta t) + (1 - \alpha)\,\theta_t^a, \qquad \alpha = 0.98. \tag{6}$$

If $\|\mathbf{g}_0 - (\mathbf{g}_0 \cdot \mathbf{u})\mathbf{u}\| < 0.3$, gravity carries no tilt information; the estimate becomes gyroscope-only with a slow leak toward neutral and is flagged.

### *G. Repetitions and Smoothness*

The tilt is sign-normalised so that repetitions are positive excursions from the neutral pose. Peaks must have a prominence of at least $\max(A_{\min}, 0.35\,(P_{95} - P_5))$ and lie at least $T_{\min}$ apart ($A_{\min} = 10°$ and $T_{\min} = 0.8$ s for repetitions; $8°$ and 0.6 s for gait cycles). A repetition spans the troughs around its peak, trimmed to where the trace leaves and regains within 5% of the trough level, and is accepted only if it returns by at least half its rise; its range is the ROM proxy, and phases are labelled at 10%, 30% and 90% of its amplitude. Smoothness is the spectral arc length [21] of the angular-speed profile,

$$\mathrm{SPARC} = -\int_0^{\omega_c} \sqrt{\Big(\frac{1}{\omega_c}\Big)^2 + \Big(\frac{d\hat{V}(\omega)}{d\omega}\Big)^2}\, d\omega, \tag{7}$$

where $\hat{V}$ is the magnitude spectrum normalised to its peak and $\omega_c \leq 10$ Hz is the highest frequency with $\hat{V} \geq 0.05$.

### *H. Bilateral Asymmetry*

Left and right are compared through normalised differences

$$d(L, R) = |L - R| / \max(|L|, |R|) \in [0, 1] \tag{8}$$

of dynamic acceleration RMS, angular-velocity RMS and 95th percentile, and ROM proxy, together with waveform dissimilarity ($1-$ peak cross-correlation) and timing. With per-leg period $P$ and cross-correlation lag $\ell$, the folded lag $\varphi = |((\ell + P/2) \bmod P) - P/2|$ gives a timing component $\varphi/(P/2)$ for synchronous exercises (squats) and $|\varphi - P/2|/(P/2)$ for alternating ones (walking). $A$ is the mean of the available components (those below their noise floor are dropped); if either side is less than 90% available the window returns SINGLE_SIDE, never a mirrored score. Its confidence is $\min(1, T/4\,\mathrm{s}) \cdot \min_s \alpha_s \cdot \sqrt{n_A/6}$ for window length $T$, side availability $\alpha_s$ and $n_A$ available components.

### *I. Movement Quality Index*

The MQI combines six components $c_i \in [0, 1]$: symmetry $(1 - A)$, ROM proxy relative to the patient's baseline (capped at 1), temporal consistency ($1-$ coefficient of variation of repetition durations), smoothness (median SPARC mapped linearly from $[-7, -1.5]$ to $[0, 1]$), force-peak consistency, and the mean pairwise correlation of time-normalised repetition profiles:

$$\mathrm{MQI} = 100\, \textstyle\sum_{i \in \mathcal{A}} w_i c_i \big/ \sum_{i \in \mathcal{A}} w_i, \tag{9}$$

over the available set $\mathcal{A}$ and at least three repetitions. With no labels to fit them, the weights are equal—an assumption, not a finding. Its confidence is $\sqrt{|\mathcal{A}|/6} \cdot \min(1, n_{\mathrm{rep}}/8) \cdot q_{\mathrm{stream}} \cdot (0.4 + 0.6\, q_{\mathrm{cal}})$, reported as LOW_CONFIDENCE below 0.5. The definition is frozen by a SHA-256 hash, and MQIs from different component sets are never compared.

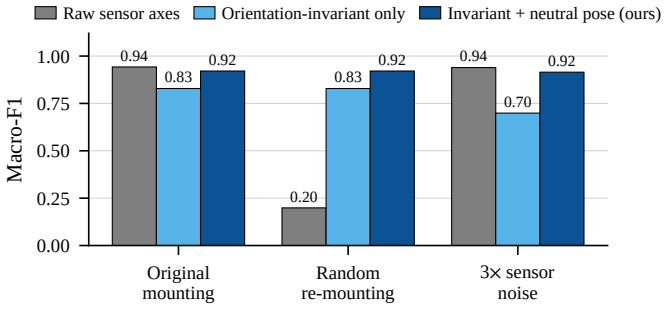



Fig. 3. Feature-set ablation (bilateral RF, 2 s windows, subject-independent macro-F1) under the original mounting, random re-mounting and 3× sensor noise.

TABLE III
SUBJECT-INDEPENDENT BENCHMARK (DAILY & SPORTS, 9 CLASSES)

| Model (features) | $W$ (s) | F1 | Folds $\mu \pm \sigma$ | Rot. | 3×noise | p95 (ms) |
|---|---|---|---|---|---|---|
| RF (raw axes, control) | 2 | 0.943 | $0.940 \pm 0.019$ | **0.199** | 0.939 | 5.4 |
| RF (invariant only) | 2 | 0.829 | $0.824 \pm 0.049$ | 0.829 | 0.699 | 5.0 |
| **RF (inv. + neutral)**$^\star$ | **2** | **0.921** | $\mathbf{0.907 \pm 0.061}$ | **0.921** | **0.915** | **12.9** |
| RF (inv. + neutral) | 3 | 0.929 | $0.916 \pm 0.063$ | 0.929 | 0.919 | 5.3 |
| RF (+ device aug.) | 2 | 0.921 | $0.908 \pm 0.061$ | 0.921 | 0.919 | 6.2 |
| HGB (inv. + neutral) | 3 | 0.945 | $0.935 \pm 0.069$ | 0.945 | 0.852 | 156.4 |
| 1D CNN | 2 | 0.871 | $0.863 \pm 0.045$ | 0.871 | 0.868 | 0.4 |
| CNN-LSTM | 2 | 0.896 | $0.892 \pm 0.030$ | 0.896 | 0.893 | 0.9 |
| CNN, pretrained | 2 | 0.898 | $0.885 \pm 0.027$ | 0.898 | 0.900 | 0.3 |
| RF, single IMU | 2 | 0.895 | $0.894 \pm 0.017$ | 0.895 | 0.859 | 5.0 |

$^\star$Deployed. F1: pooled out-of-fold macro-F1; Folds: per-fold macro-F1 (4 folds); Rot.: after random sensor rotation. Pretrained: encoder pretrained on PAMAP2. Single IMU: each Daily & Sports leg plus the PAMAP2 ankle. Transfer: Daily & Sports→PAMAP2 0.241, reverse 0.153.

## V. EXPERIMENTAL SETUP

**Datasets.** *Daily and Sports Activities* [22] provides 8 subjects with Xsens units on both legs *recorded simultaneously* and trains the bilateral model (60,424 windows); *PAMAP2* [23] adds 9 subjects with an ankle IMU to the single-IMU model and encoder pretraining; *UCI HAR* [24] (561 precomputed features) is a tooling check (macro-F1: RF 0.923, HGB 0.936). No single-IMU recording is mirrored into a left–right pair, and one standing segment per subject emulates calibration and is excluded from evaluation.

**Protocol.** Splits are GroupKFold(4) by subject, with windows cut inside recordings, normalisation fitted on training subjects only and out-of-fold predictions pooled; macro-F1 is the primary metric [7]. Perturbations are applied to raw test windows: a uniformly random rotation (to the signals and to $\mathbf{g}_0$), Gaussian noise of 0.03 g and 1.5°/s (3×), per-axis gyroscope bias from $U(-3, 3)$°/s, accelerometer gain from $U(0.95, 1.05)$ and a 5% time stretch. Transfer is tested between the Daily and Sports legs and the PAMAP2 ankle. Latency is single-window features plus prediction on one laptop CPU thread (200 runs).

**Verification.** Public windows are replayed as quantised, jittered 100 Hz device streams; a simulator straps each IMU at a random orientation and generates five exercises with known ROM, left-side deficits (0–0.6) and faults.

## VI. RESULTS AND DISCUSSION

### A. Mounting Invariance

Fig. 3 confirms the proposition empirically. Raw-axis features score highest on the original mounting (0.943) but collapse to 0.199 once the test sensors are rotated, as a re-attached strap does. Invariant channels alone are immune to rotation but lose posture (0.829) and are fragile under noise (0.699); adding neutral-pose posture restores 0.921, unchanged under rotation and nearly so under 3× noise (0.915).

### B. Model Selection

Table III reports the benchmark. HGB is most accurate (0.945 at 3 s) but needs 156 ms per window, three times the budget. The RF at 2 s (0.921) lies within 0.01 of its family's best and responds faster, so rule (5) selects it; the augmented RF ties and loses on size (15.4 vs. 9.6 MB) despite better noise robustness. Per-fold scores spread by about ±0.06 (one difficult held-out subject group scores 0.831), so the 0.024 gap between HGB and RF is smaller than the fold-to-fold variation and the latency constraint decides. The deep models trail by 0.023–0.051, but PAMAP2 pretraining lifts the CNN by 0.027 with less than half the RF's fold variability, keeping it the candidate for force fusion once device data exist. On the server the deployed model needs 4.7 ms per window (median), but its 220 MB footprint rules out on-device inference without a smaller model.

### C. Per-Class Behaviour and Selective Prediction

On the 60,424 out-of-fold windows, cycling, running, standing and lying exceed 0.99 recall and other exercise reaches 0.98. A quarter of sitting windows are labelled lying, the posture pair one segment per leg separates least well, and walking and stair windows drift toward *other exercise*, which includes stepper and cross-trainer movements. At $\tau = 0.3$ the model answers 99.99% of windows with 91.8% accuracy; $\tau = 0.7$ gives 97.3% on 80.2% of windows. The threshold must be re-derived on device data.

### D. Domain Gap and Train–Serve Parity

Cross-dataset transfer collapses to 0.241 and 0.153 macro-F1, measuring what a change of sensor and placement costs, so public-data scores are pretraining evidence, not device accuracy. The device path itself is faithful: replay reproduces the training-path prediction for 98.4% of 800 windows, and the full processor agrees with the labels on 95.6% of 1,152 windows (integrity checks on subjects seen in training, not generalisation estimates).

### E. Movement Metrics in Simulation

Against simulated ground truth, the tilt is within 2.5° (median) under random strap orientations, repetition counts fall within 1–3 of the simulated cycles (incomplete cycles excluded), and all four injected faults (packet loss, IMU dropout, frozen IMU, impossible values) are detected. The ROM proxy is within ±1.5% for the four repetitive exercises but 9.4% low for walking, plausibly because heel strikes

violate the quasi-static assumption behind $\theta_t^a$. As the injected deficit grows from 0 to 0.6, the squat asymmetry score rises monotonically from 0.01 to 0.26 (walking 0.08 to 0.40) and the MQI falls from 96.0 to 91.0, 88.5 and 80.8, with a repeatability of $\pm 0.08$ over ten seeds. Removing the right IMU changes the MQI from 96.0 to 95.2 only because the symmetry term disappears—hence MQIs from different component sets are never compared.

## VII. Limitations and Future Work

RehabSense is a research prototype, not a medical device. Its activity model is validated on public data only; no physical-device recording has yet been evaluated, and the transfer results predict a drop until the model is adapted. The ROM proxy is a segment tilt, not a knee angle; the force channels are an uncalibrated load proxy; and the MQI weights and SPARC bounds are assumptions characterised only in simulation. The next stage is an ethics-approved pilot: consented participants recorded on two days with re-strapping, therapist labels, zero-shot then fine-tuned leave-one-subject-out evaluation, goniometer comparison and MQI reliability (intraclass correlation). By lowering the cost of objective home monitoring, the work supports UN Sustainable Development Goals 3 and 9.

## VIII. Conclusion

RehabSense pairs a US$18 bilateral wearable with an analytics pipeline that is robust to how patients put it on and explicit about what its numbers mean. Measuring posture against a calibrated neutral pose reconciles provable orientation invariance with posture recognition (macro-F1 0.921, unchanged under re-mounting, versus 0.199 for raw axes); the platform is now ready for its on-device pilot study.


## Acknowledgment

The authors thank the Department of Computer Engineering, Vishwakarma Institute of Technology, Pune, for its support, and the creators of the public datasets used here.


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